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Wireless out-of-band management of information technology assets

Abstract

An apparatus comprises two or more connection ports, at least one wireless network interface, and at least one processing device comprising a processor coupled to a memory. The at least one processing device is configured to receive, utilizing a first one of the two or more connection ports, a first type of out-of-band management traffic from a managed information technology asset, to receive, utilizing a second one of the two or more connection ports, a second type of out-of-band management traffic from the managed information technology asset, and to communicate, via the at least one wireless network interface, the first and second types of out-of-band management traffic to one or more additional processing devices.

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USPC: 370/229; 709/223

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10785652	12/2019	Ravindranath	N/A	H04W 12/48
11233692	12/2021	Narayanan et al.	N/A	N/A
2003/0101239	12/2002	Ishizaki	N/A	N/A
2007/0206630	12/2006	Bird	370/465	H04L 41/24
2008/0072291	12/2007	Carley	N/A	N/A
2010/0281094	12/2009	Holt	709/201	H04L 41/344
2014/0204955	12/2013	Kelly	N/A	N/A
2014/0362859	12/2013	Addanki et al.	N/A	N/A
2017/0230251	12/2016	Dube	N/A	H04L 67/1097
2017/0244640	12/2016	Lin et al.	N/A	N/A
2021/0044563	12/2020	Reyes et al.	N/A	N/A
2022/0116260	12/2021	Narayanan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2012094129	12/2011	WO	H04W 24/10

OTHER PUBLICATIONS

G. Rieger, “socat (1)—Linux Man Page,” https://linux.die.net/man/1/socat, Accessed Jul. 15, 2022, 33 pages. cited by applicant

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Background/Summary

FIELD

(1) The field relates generally to information processing, and more particularly to management of information processing systems.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. Information handling systems may be used to process, compile, store and communicate various types of information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary (e.g., in what information is handled, how the information is handled, how much information is processed, stored, or communicated, how quickly and efficiently the information may be processed, stored, or communicated, etc.). Information handling systems may be configured as general purpose, or as special purpose configured for one or more specific users or use cases (e.g., financial transaction processing, airline reservations, enterprise data storage, global communications, etc.). Information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

SUMMARY

(3) Illustrative embodiments of the present disclosure provide techniques for wireless out-of-band management of information technology assets.

(4) In one embodiment, an apparatus comprises two or more connection ports, at least one wireless network interface, and at least one processing device comprising a processor coupled to a memory. The at least one processing device is configured to receive, utilizing a first one of the two or more connection ports, a first type of out-of-band management traffic from a managed information technology asset, to receive, utilizing a second one of the two or more connection ports, a second type of out-of-band management traffic from the managed information technology asset, and to communicate, via the at least one wireless network interface, the first and second types of out-of-band management traffic to one or more additional processing devices.

(5) These and other illustrative embodiments include, without limitation, methods, apparatus, networks, systems and processor-readable storage media.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram of an information processing system configured for wireless out-of-band management of information technology assets in an illustrative embodiment.

(2) FIG. **2** shows an information processing system including a power over ethernet powered wireless out-of-band management device configured for connection to a network switch with horizontally aligned management and console ports in an illustrative embodiment.

(3) FIG. **3** shows an information processing system including a power over ethernet powered wireless out-of-band management device configured for connection to a network switch with vertically aligned management and console ports in an illustrative embodiment.

(4) FIG. **4** shows functional blocks of a power over ethernet powered wireless out-of-band management device in an illustrative embodiment.

(5) FIG. **5** shows an information processing system including a universal serial bus powered wireless out-of-band management device configured for connection to a network switch with horizontally aligned management, console and universal serial bus ports in an illustrative embodiment.

(6) FIG. **6** shows an information processing system including a universal serial bus powered

wireless out-of-band management device configured for connection to a network switch with vertically aligned management and console ports in an illustrative embodiment.
(7) FIG. **7** shows functional blocks of a universal serial bus powered wireless out-of-band management device in an illustrative embodiment.
(8) FIG. **8** is a flow diagram of an exemplary process for wireless out-of-band management of information technology assets in an illustrative embodiment.
(9) FIGS. **9** and **10** show examples of processing platforms that may be utilized to implement at least a portion of an information processing system in illustrative embodiments.

DETAILED DESCRIPTION

(10) Illustrative embodiments will be described herein with reference to exemplary information processing systems and associated computers, servers, storage devices and other processing devices. It is to be appreciated, however, that embodiments are not restricted to use with the particular illustrative system and device configurations shown. Accordingly, the term “information processing system” as used herein is intended to be broadly construed, so as to encompass, for example, processing systems comprising cloud computing and storage systems, as well as other types of processing systems comprising various combinations of physical and virtual processing resources. An information processing system may therefore comprise, for example, at least one data center or other type of cloud-based system that includes one or more clouds hosting tenants that access cloud resources.
(11) FIG. **1** shows an information processing system **100** configured in accordance with an illustrative embodiment. The information processing system **100** is assumed to be built on at least one processing platform and provides functionality for wireless out-of-band management (WOOBM) of information technology (IT) assets of an IT infrastructure **105**, such as IT asset **106**, using a WOOBM device **110**. The WOOBM device **110** is coupled, via one or more networks **104**, to a client device **102** and an IT support system **108**. Out-of-band management (OOBM) for the IT asset **106** refers to management of the IT asset **106** that is at least partially under the control or direction of another device (e.g., an “out-of-band” device such as WOOBM device **110**, the IT support system **108**, via a management network that includes one or both of the WOOBM device **110** and the IT support system **108**, etc.), as compared with management of the IT asset **106** that is entirely under the control or direction of the IT asset **106** itself (e.g., “in band”). OOBM may include servicing, configuration and other management actions that are controlled or directed by an “out-of-band” device.
(12) The IT asset **106** may include one or more physical and/or virtual computing resources. Physical computing resources may include physical hardware such as servers, storage systems, networking equipment, Internet of Things (IoT) devices, other types of processing and computing devices including desktops, laptops, tablets, smartphones, etc. Virtual computing resources may include virtual machines (VMs), containers, etc. In various embodiments described herein, it is assumed that the IT asset **106** comprises a network switch that is managed by the client device **102** and/or IT support system **108** utilizing the WOOBM device **110**.
(13) The WOOBM device **110** comprises a processor **112**, a memory **114**, one or more wireless network interfaces **116** and connection ports **118-1**. Two or more of the connection ports **118-1** of the WOOBM device **110** are configured for mating with OOBM connection ports **118-2** of the IT asset **106**, possibly via a WOOBM connection port adapter **119** having corresponding connectors **118-3**. In some embodiments, the OOBM connection ports **118-2** of the IT asset **106** a management port and a console port. At least one of the connection ports **118-1** of the WOOBM device **110** are configured for mating with corresponding connection ports **118-4** of the client device **102** via cabling **111**. The connection ports **118-4** of the client device **102** may include, for example, flash drive or debug console ports, Universal Serial Bus (USB) ports, etc.
(14) It should be noted that the coupling of the client device **102** to the WOOBM device **110** via cabling **111** is optional. The client device **102** may also or alternatively be coupled to the WOOBM

device **110** via the one or more wireless networks **104**, such as cellular (e.g., Long Term Evolution (LTE), 5G, etc.) networks, WiFi (e.g., WiFi 6) networks, other short-range wireless networks (e.g., Bluetooth, Bluetooth Low Energy (BLE), Near Field Communication (NFC), etc.). The WOOBM device **110** is also configured for coupling via the wireless networks **104** to the IT support system **108**. The wireless networks **104** coupling the IT support system **108** to the WOOBM device **110** may be the same as or different than those used for coupling the client device **102** to the IT support system **108**.

(15) The IT support system **108** may be operated by an enterprise or other entity that provides support services for the IT infrastructure **105**. The enterprise or other entity that provides support services for the IT infrastructure **105** may be the same as or different than the enterprise or other entity that operates the IT infrastructure **105**. For example, an enterprise or other entity operating the IT infrastructure **105** may subscribe to or otherwise utilize the IT support system **108** for providing support or management services for the IT asset **106**. In some cases, the IT support system **108** is operated by a vendor of the IT asset **106**. As used herein, the term “enterprise system” is intended to be construed broadly to include any group of systems or other computing devices. For example, the IT asset **106** of the IT infrastructure **105** may provide a portion of one or more enterprise systems. A given enterprise system may also or alternatively include the client device **102** and/or the IT support system **108**. In some embodiments, an enterprise system includes one or more data centers, cloud infrastructure comprising one or more clouds, etc. A given enterprise system, such as cloud infrastructure, may host assets that are associated with multiple enterprises (e.g., two or more different business, organizations or other entities).

(16) The client device **102** may comprise, for example, physical computing devices such as IoT devices, mobile telephones, laptop computers, tablet computers, desktop computers or other types of devices utilized by members of an enterprise, in any combination. Such devices are examples of what are more generally referred to herein as “processing devices.” Some of these processing devices are also generally referred to herein as “computers.” The client device **102** may also or alternately comprise virtualized computing resources, such as VMs, containers, etc.

(17) The client device **102** in some embodiments comprises a computer associated with a particular company, organization or other enterprise. Thus, the client device **102** may be considered an example of an asset of an enterprise system. In addition, at least portions of the system **100** may also be referred to herein as collectively comprising one or more “enterprises.” Numerous other operating scenarios involving a wide variety of different types and arrangements of processing nodes are possible, as will be appreciated by those skilled in the art.

(18) Although not explicitly shown in FIG. **1**, one or more input-output devices such as keyboards, displays or other types of input-output devices may be used to support one or more user interfaces to the client device **102**, the IT support system **108**, the WOOBM device **110**, and the IT asset **106**, as well as to support communication between these elements and other related systems and devices not explicitly shown.

(19) The client device **102** is configured to access or otherwise utilize the IT infrastructure **105**. In some embodiments, the client device **102** is assumed to be associated with a system administrator, IT manager or other authorized personnel responsible for managing the IT asset **106** of the IT infrastructure **105** (e.g., where such management includes performing servicing of the IT asset **106**, or of applications or other software that runs on the IT asset **106**). The IT support system **108** may also be used to provide management or other support for the IT asset **106** of the IT infrastructure. In some embodiments, the IT asset **106** of the IT infrastructure **105** is owned or operated by the same enterprise that operates the client device **102** and/or the IT support system **108** (e.g., where an enterprise such as a business provides support for the assets it operates). In other embodiments, the IT asset **106** of the IT infrastructure **105** may be owned or operated by one or more enterprises different than the enterprise which operates the client device **102** and/or the IT support system **108** (e.g., a first enterprise provides support for assets that are owned by multiple different customers,

business, etc.). Various other examples are possible.

(20) In some embodiments, the client device **102** and/or IT asset **106** may implement host agents that are configured for automated transmission of information that is to be provided to the IT support system **108**. Such host agents may also or alternatively be configured to automatically receive from the IT support system **108** management or configuration commands or other information. It should be noted that a "host agent" as this term is generally used herein may comprise an automated entity, such as a software entity running on a processing device. Accordingly, a host agent need not be a human entity.

(21) The client device **102**, IT support system **108**, WOOBM device **110**, IT asset **106** and other elements of the information processing system **100** in the FIG. **1** embodiment are assumed to be implemented using at least one processing device. Each such processing device generally comprises at least one processor and an associated memory, and implements one or more functional modules or logic for controlling certain features of the information processing system **100**. In the FIG. **1** embodiment, for example, the WOOBM device **110** comprises a processor **112** and memory **114** that implements OOBM traffic management logic **120**. At least portions of the OOBM traffic management logic **120** may be implemented at least in part in the form of software that is stored in memory **114** and executed by processor **112**. The OOBM traffic management logic **120** is configured to receive, utilizing a first one of the connection ports **118-1**, a first type of OOBM traffic from managed IT asset **106**. The OOBM traffic management logic **120** is also configured to receive, utilizing a second one of the connection ports **118-2**, a second type of OOBM traffic from the managed IT asset **106**. The OOBM traffic management logic **120** is further configured to communicate, via the wireless network interfaces **116**, the first and second types of OOBM traffic to one or more additional processing devices (e.g., client device **102**, IT support system **108**). The OOBM traffic management logic **120** may be further configured to receive, via the wireless network interfaces **116**, additional OOBM traffic from at least one of the one or more additional processing devices, and to separate the additional OOBM traffic into the first type of OOBM traffic and the second type of OOBM traffic. The first type of OOBM traffic in the additional OOBM traffic may be communicated utilizing the first connection port to a first one of the OOBM connection ports **118-2** of the managed IT asset **106**, and the second type of OOBM traffic in the additional OOBM traffic may be communicated utilizing the second connection port to a second one of the OOBM connection ports **118-2** of the managed IT asset **106**.

(22) It is to be appreciated that the particular arrangement of the client device **102**, the IT infrastructure **105**, the IT support system **108**, the WOOBM device **110**, and the WOOBM connection port adapter **119** illustrated in the FIG. **1** embodiment is presented by way of example only, and alternative arrangements can be used in other embodiments. As discussed above, for example, the WOOBM connection port adapter **119** is optional. In some cases, the connection ports **118-1** of the WOOBM device **110** are coupled to the OOBM connection ports **118-2** of the IT asset **106** directly or via cabling rather than via the WOOBM connection port adapter **119**.

(23) The IT support system **108** and other portions of the system **100**, as will be described in further detail below, may be part of cloud infrastructure.

(24) The client device **102**, the IT infrastructure **105**, the IT support system **108**, the WOOBM device **110** and other components of the information processing system **100** in the FIG. **1** embodiment are assumed to be implemented using at least one processing platform comprising one or more processing devices each having a processor coupled to a memory. Such processing devices can illustratively include particular arrangements of compute, storage and network resources.

(25) The client device **102**, the IT infrastructure **105**, the IT support system **108**, the WOOBM device **110**, or components thereof or other components of the information processing system **100** in the FIG. **1** embodiment, may be implemented on respective distinct processing platforms, although numerous other arrangements are possible. For example, in some embodiments at least portions of the client device **102** and the IT support system **108** are implemented on the same

processing platform.
(26) The term “processing platform” as used herein is intended to be broadly construed so as to encompass, by way of illustration and without limitation, multiple sets of processing devices and associated storage systems that are configured to communicate over one or more networks. For example, distributed implementations of the system **100** are possible, in which certain components of the system reside in one data center in a first geographic location while other components of the system reside in one or more other data centers in one or more other geographic locations that are potentially remote from the first geographic location. Thus, it is possible in some implementations of the system **100** for client device **102**, the IT infrastructure **105**, the IT support system **108**, the WOOBM device **110**, or portions or components thereof, to reside in different data centers. Numerous other distributed implementations are possible.
(27) Additional examples of processing platforms utilized to implement components of the system **100** in illustrative embodiments will be described in more detail below in conjunction with FIGS. **9** and **10**.
(28) It is to be appreciated that these and other features of illustrative embodiments are presented by way of example only, and should not be construed as limiting in any way.
(29) It is to be understood that the particular set of elements shown in FIG. **1** for WOOBM of IT assets is presented by way of illustrative example only, and in other embodiments additional or alternative elements may be used. Thus, another embodiment may include additional or alternative systems, devices and other network entities, as well as different arrangements of modules and other components.
(30) It is to be appreciated that these and other features of illustrative embodiments are presented by way of example only, and should not be construed as limiting in any way.
(31) OOBM data may originate from multiple different ports of a managed IT asset (e.g, a network switch), such as serial data from a console port and ethernet traffic from a management port. Managing such OOBM data or traffic using two different networks is cumbersome and inefficient. Conventional approaches have attempted to provide a remedy utilizing processors that provide Serial-Over-LAN (SOL) access. Such conventional approaches, however, suffer from various technical problems and challenges, including the need to set up and manage a dynamic host control protocol (DHCP) server as part of a management network, distributing assigned SOL IP addresses, etc. Further, not all existing network switches or other IT assets support SOL in hardware. Accordingly, there is a need for technical solutions for accessing OOBM ports of a managed IT asset to perform support of management of that managed IT asset. This may include using unified management server ports on an OOBM device to handle or reach both console and management traffic.
(32) 5G evolution and deployments are happening at a rapid pace, and are transforming the way networks are being designed. Cloud operators may offer “Private 5G” fully managed services for the enterprise and edge. With the deployment of 5G, the use of disaggregated network switches in these environments presents new opportunities as well as challenges. Various IT assets, including network switches, support OOBM with multiple ports (e.g., a console port and an ethernet or other management port). Hyperscalers have realized capital expenditure (CapEx) and operating expense (OpEx) benefits of consolidating ethernet and console management networks. With private 5G, there are clear synergistic benefits in integrating the OOBM access of the network switches into the radio access network (RAN) itself.
(33) There are various technical challenges associated with wired OOBM usage (e.g., for both ethernet and console OOBM). In some environments, console connectivity is available only on request (e.g., console servers are not deployed to cut cost and/or cabling overhead). If units do not respond over a management network (e.g., over an ethernet port), OOBM operations must wait for a customer's data center personnel to get physical access and schedule debug sessions (e.g., facilitated by connecting a laptop's COM port). Further, in many cases (e.g., file system and/or

partition corruptions, SSD failures, etc.), the managed IT asset may be stuck in its basic input/output system (BIOS). In such cases, the only way to jump-start the managed IT asset (e.g., to avoid return merchandise authorization (RMA)) would be using a bootable universal serial bus (USB) (e.g., with Open Network Install Environment (ONIE) or Diagnostic Operating System (DIAG-OS) capabilities), which requires a technician to have physical access to the console of the managed IT asset.

(34) Additional technical challenges relate to customers or end-users (e.g., including large enterprises and hyperscalers) wanting to focus on their applications or business logic rather than component firmware upgrades (e.g., for BIOS, complex programmable logic devices (CPLDs), field-programmable gate arrays (FPGAs), baseboard management controllers (BMCs), etc.). Even if upgrade procedures are well documented (e.g., using ONIE updater, vendor diagnostics, network operating software (NOS), etc.), customers or other end-users seek to avoid the hassle of integrating such upgrade procedures into their automation. Further, some non-sophisticated customers or other end-users may fail to correctly integrate such upgrade procedures into their automation.

(35) Further technical challenges relate to the managed IT assets themselves, which may spend an (unpredictably) variable time in partner/depot facilities, such that the firmware of the IT assets may not be current when the IT assets are onboarded or zero touch provisioned (ZTP'd) in a customer or end user data center. The cost of upgrading components in second touch facilities may also be prohibitively expensive (e.g., potentially negative margin scenarios).

(36) The concepts of Preboot Execution Environment (PXE) over wireless, configuration, control, and upgrade over a management wireless uplink, Firmware Over-the-Air (FOTA), etc., have been used for wireless access points (APs). These options, however, have been out of reach for certain types of IT assets such as network switches. With the rollout of enterprise 5G, as well as the adoption of WiFi 6, a solution that provides a multiplexed uplink to both management (e.g., ethernet) and console ports with a single device can conveniently address these and other technical challenges. Illustrative embodiments provide a WOOBM solution that utilizes a WOOBM device (also referred to as a dongle) that derives power from a managed IT asset (e.g., a network switch) and uplinks to the RAN (e.g., of a 5G or other cellular network), to WiFi access points, etc. In some embodiments, the WOOBM device derives power from one of the OOBM ports of the managed IT asset (e.g., the management port of the IT asset, where the management port comprises a power over ethernet (PoE) port). In other embodiments, the WOOBM device derives power from a non-OOBM port of the managed IT asset, such as a USB port that is distinct from OOBM ports of the managed IT asset.

(37) In the description herein, connections between components or systems within the figures are not intended to be limited to direct connections unless otherwise specified. Rather, data between these components may be modified, re-formatted, or otherwise changed by intermediary components. Also, additional or fewer connections may be used. It shall also be noted that the terms “coupled,” “connected,” or “communicatively coupled” shall be understood to include direct connections, indirect connections through one or more intermediary devices, and wireless connections.

(38) It should be appreciated that although certain embodiments are described herein within the context of certain connectors and ports, embodiments are not so limited. For example, while various embodiments are described with respect to console and management port RJ45 jacks, it should be understood that various other types of connectors or ports (e.g., universal serial bus (USB), micro-USB, etc.) may be used instead of or in addition to RJ45 jacks. The terms “managed switch port,” and “management switch port” and “management port” are used interchangeably. Similarly, “PoE OOBM switch” and “OOBM switch” are used interchangeably. “OOBM switch port” refers to the port of an OOBM switch.

(39) Vendors of IT assets, such as network switches, may provide OOBM ports (e.g., RJ45 ports)

that are co-located—a console port and a management ethernet port (e.g., a 1G port). These OOBM ports may be aligned horizontally (e.g., as shown in FIGS. **2** and **5**) or vertically (e.g., as shown in FIGS. **3** and **6**). The management ethernet port on the network switch or other IT asset being managed may be made 802.1af PoE capable to power a WOOBM device or dongle that may be attached to the network switch or other IT asset being managed (e.g., possibly using a WOOBM connection port adapter).
(40) FIG. **2** shows a system **200** including a WOOBM device **210** and matching WOOBM connection port adapter **219** that have horizontally aligned ports. The WOOBM device **210** is configured to provide connectivity between a network switch **206** with horizontally aligned OOBM ports, including a management ethernet port **218-2-1** and a console port **218-2-2**. The WOOBM device **210** includes an ethernet port **218-1-1**, a serial port **218-1-2**, and a flash drive/debug console **218-1-3**. The WOOBM device **210** includes adjustable antennas **217** which connect to an IT support system **208** over one or more wireless networks **204**, shown in FIG. **2** as a radio access network (RAN) and next generation core network (NGCN) (e.g., of a private 5G enterprise core network, which may in turn have the ability to access secure remote services or support assistance services offered by that enterprise). The flash drive/debug console **218-1-3** may be used to debug software running on the WOOBM device **210**. The flash drive/debug console **218-1-3** may use a 4-pin connector or any other desired type of debugging connection or mechanism.
(41) The network switch **206**, also referred to as a managed switch **206**, comprises the management ethernet port **218-2-1**, also referred to simple as management port **218-2-1**, and a console port **218-2-2** that are co-located. The management port **218-2-1** is PoE capable to provide power to the WOOBM device **210**. In the FIG. **2** example, the management port **218-2-1** and console port **218-2-2** are co-located on the managed switch **206** (e.g., next to other ports), which may be on the front or rear of the managed switch **206**. The management port **218-2-1** and console port **218-2-2** may be connected to internal components (e.g., one or more application-specific integrated circuits (ASICs) or other processors, not shown) of the managed switch **206**.
(42) The WOOBM connection port adapter **219**, also referred to as a WOOBM adapter **219**, is an optional device that may be used to connect the WOOBM device **210** to the managed switch **206**. The WOOBM adapter **219** may include sets of ethernet connectors **218-3-1** for connection to ethernet port **218-1-1** of the WOOBM device **210** and the management port **218-2-1** of the managed switch **206**, as well as serial connectors **218-3-2** for connection to serial port **218-1-2** of the WOOBM device **210** and the console port **218-2-2** of the managed switch **206**. In other embodiments, the management port **218-2-1** and ethernet port **218-1-1**, as well as the serial port **218-1-2** and console port **218-2-2**, may be connected without utilizing WOOBM adapter **219**, such as using Ethernet and/or console cables. The WOOBM adapter **219**, however, may be preferred in implementations in which the WOOBM device **210** has a certain length (e.g., greater than 3 inches (in)) and/or a considerable cross-section that may potentially obstruct air flow (e.g., when being suspended from a rack with no support, the WOOBM adapter **219** may be used to position the WOOBM device **210** in the management port **218-2-1** and console port **218-2-2** of the managed switch **206**, preventing the WOOBM device **210** from negatively impacting cooling).
(43) The distances between the ethernet connectors **218-3-1** and the serial connectors **218-3-2** (e.g., the distances between midpoints of the male connectors) that attach to the WOOBM device **210** and managed switch **206** may be fixed and substantially correspond to the distances between the jacks or ports **218-1-1** and **218-1-2** of the WOOBM device **210** and the jacks or ports **218-2-1** and **218-2-2** of the managed switch **206**. In practice, the distances between the management and console ports of managed switches or other IT assets may vary (e.g., across different vendors, across different models or product lines from the same or different vendors, etc.). Accordingly, the distances between the ethernet connectors **218-3-1** and serial connectors **218-3-2** on both sides of the WOOBM adapter **219** may be customized to match the dimensions of corresponding ports on both the WOOBM device **210** and the managed switch **206**. Assuming a user desires integrated

management and console access for an IT asset (e.g., management switch **206**) having horizontally-aligned management and console ports, the user may select a particular one of multiple WOOBM adapters (e.g., having different spacing between their ethernet and serial connectors) that matches that IT asset as well as the WOOBM device **210** used by that user. In some embodiments, the WOOBM adapter **219** and/or WOOBM device **210** may comprise unique identifiers such as stock keeping units (SKUs) that simplify matching of the WOOBM device **210** to the WOOBM adapter **219**. It should be understood that many SKUs or other identifiers may exist for different combinations of WOOBM devices and mating WOOBM adapters.

(44) In some embodiments, the ethernet connectors **218-3-1** and serial connectors **218-3-2** of the WOOBM adapter **219** are designed to extend into the WOOBM device **210** by a certain length to support the weight of the WOOBM device **210**. Various other supporting structures may also or alternatively be used such that the WOOBM device **210** can be securely attached to a rack (not shown).

(45) The WOOBM device **210** may be configured for communication with a client device **202** running a WOOBM application **203** (e.g., for use in controlling or access the WOOBM device **210** and/or the managed switch **206** that the WOOBM device **210** is connected to). The WOOBM application **203** may be used for debugging of the WOOBM device **210** (e.g., via flash drive/debug console **218-1-3**, via a wireless network connection such as a Bluetooth or BLE connection).

(46) The management and console ports of a managed switch or other IT asset may be aligned horizontally or vertically. If the management and console ports are aligned horizontally (e.g., along a common horizontal axis as shown in the managed switch **206** of FIG. **2**), then the horizontally-aligned WOOBM device **210** and WOOBM adapter **219** may be used. If, however, the management and console ports are aligned vertically (e.g., along a common vertical axis as shown in the managed switch **306** of the system **300** of FIG. **3**), then a vertically-aligned WOOBM device **310** and WOOBM adapter **319** may be used. The WOOBM device **310**, similar to WOOBM device **210**, includes an ethernet port **318-1-1**, a serial port **318-1-2** and a flash drive/debug console **318-1-3**, as well as antennas **317** for connecting with an IT support system **308** via RAN and NGCN **304**. The managed switch **306**, similar to managed switch **206**, includes a management ethernet port **318-2-1** (e.g., a PoE port) and a console port **318-2-2**. The management ethernet port **318-2-1**, also referred to as management port **318-2-1**, and console port **318-2-2** of the managed switch **306** are vertically aligned, as compared with the horizontally aligned management port **218-2-1** and console port **218-2-2** of the managed switch **206**.

(47) It should be noted that RJ45 ports are depicted in FIGS. **2** and **3** for purposes of illustration only. While RJ45 jacks or ports have become the de facto standard for network switches, various other types of ports or connectors may be used as desired for network switches and other types of IT assets. Further, while FIGS. **2** and **3** show examples of managed switches **206** and **306** with horizontally and vertically aligned management and console ports, respectively, various other arrangements of management and console ports may be used in other network switches and other types of IT assets (e.g., such as diagonally juxtaposed management and console ports, management and console ports which are separated by one or more other ports as described in further detail below with respect to FIGS. **5-7**, etc.).

(48) In operation, once the WOOBM device **210** or **310** is plugged into the managed switch **206** or **306** (e.g., via the WOOBM adapters **219**, **319**, or via cables coupling the management ports **218-2-1**, **318-2-1** to the ethernet ports **218-1-1**, **318-1-1** and the console ports **218-2-2**, **318-2-2** to the serial ports **218-1-1**, **318-1-2**), the WOOBM device **210** or **310** can multiplex signals from the ethernet ports **218-1-1**, **318-1-1** and the serial ports **218-1-2**, **318-1-2** and provide the resulting multiplexed signals over wireless network interfaces (e.g., antennas **217**, **317**) to respective IT support systems **208**, **308** (e.g., via RAN/NGCNs **204**, **304**).

(49) In some embodiments, the serial ports **218-1-2**, **318-1-2** and console ports **218-2-2**, **318-2-2** may handle WOOBM traffic, such as using a relatively low speed serial protocol and command

line interface (CLI) commands. A user (e.g., of client device **202**, **302** or IT support system **208**, **308**) may utilize a serial RS232 connection to access the console ports **218-2-2**, **318-2-2** (e.g., to access a BIOS of the managed switch **206**, **306**). The console ports **218-2-2**, **318-2-2** may be coupled to a reverse telnet session (e.g., on IT support systems **208**, **308**, on client devices **202**, **302**, etc.) that may act as the console server. In some cases, the WOOBM devices **210**, **310** themselves act as the console server for such reverse telnet sessions, as will be discussed in further detail below.
(50) A user (e.g., of client device **202**, **302** or IT support system **208**, **308**) may further use ethernet connectivity of the management ports **218-2-1**, **318-2-1** (e.g., which may be 1G or 10G ports running an 802.3 Ethernet protocol and having an IP address that can be reached by using a telnet session) to facilitate queries, and use Simple Network Management Protocol (SNMP) or other network management protocols to remotely manage the managed switches **206**, **306** using WOOBM. In some cases, the management ports **218-2-1**, **318-2-1** are linked to a management switch in the IT support system **208**, **308** and/or a management network within the IT support system **208**, **308** or accessible via client devices **202**, **302** using wireless network protocols. In some embodiments, console characters from multiplexed WOOBM traffic (e.g., provided from the WOOBM device **210**, **310** over wireless networks using antennas **217**, **317**) may be demultiplexed, and a reverse telnet session may be established to the console ports **218-2-2**, **318-2-2** through a management switch port of a management switch in the IT support system **208**, **308**.
(51) The WOOBM devices **210**, **310** may implement a bump-in-the-wire(less) design that consolidates management traffic (e.g., from management port **218-2-1**, **318-2-1**) and console server traffic (e.g., from console port **218-2-2**, **318-2-2**) into a single wireless uplink, by multiplexing traffic. This allows a user (e.g., of client device **202**, **302** or IT support system **208**, **308**) to access both the management ports **218-2-1**, **318-2-1** and console ports **218-2-2**, **318-2-2** of the managed switches **206**, **306** through a single WOOBM device **210**, **310** (e.g., using reverse telnet) and without having to handle two different types of network connectivity. As a result, multiplexing different types of traffic facilitates the ease of configuration and control of the managed switches **206**, **306**.
(52) In some embodiments, as discussed above, the management ports **218-2-1**, **318-2-1** are assumed to be PoE capable, such that when the WOOBM device **210**, **310** is connected thereto (e.g., using the WOOBM adapter **219**, **319** or cables), the management ports **218-2-1**, **318-2-1** act as power sourcing equipment (e.g., 802.1af PoE ports) and the WOOBM devices **210**, **310** become powered devices. Advantageously, this eliminates the need to energize the WOOBM devices **210**, **310** by using additional power lines that, in addition to increasing installation cost, may obstruct airflow. Reducing cabling and cable clutter by a significant amount further improves rack space use (e.g., in favor of providing increased computing space). It should be appreciated, however, that the WOOBM devices **210**, **310** may be energized via other mechanisms, such as through separate USB connections (e.g., as described below with respective to FIGS. **5-7**), though batteries or other power sources internal to the WOOBM devices **210**, **310**, through external power sources other than the managed switches **206**, **306**, etc.
(53) The WOOBM devices **210**, **310** enable the managed switches **206**, **306** to be configured and/or controlled using a single network operating software (NOS)/touchpoint. This advantageously simplifies the configuration and management of both the management ports **218-2-1**, **318-2-1** and console ports **218-2-2**, **318-2-2** using the WOOBM devices **210**, **310** as an integrated device. Further, the technical solutions enabled by the WOOBM devices **210**, **310** are vendor-neutral, and agnostic to the NOS, BIOS, Open Network Install Environment (ONIE), Diagnostic Operating System (DIAG-OS) software capabilities, or other software running on the managed switches **206**, **306** (or other IT assets being managed using a WOOBM device). The technical solutions enabled by the WOOBM devices **210**, **310** are further agnostic to the processors or other hardware that may be embedded in the managed switches **206**, **306** (or other IT assets being managed using a

WOOBM device). Furthermore, the technical solutions described herein can significantly reduce power consumption and operating cost.

(54) FIG. **4** shows a functional block diagram of a WOOBM device **410** (which may be the horizontally aligned WOOBM device **210** or the vertically aligned WOOBM device **310**). The WOOBM device **410** comprises an ethernet port **418-1-1** and console port **418-1-2**, both depicted as RJ45 ports, as well as a wireless network interface **416** coupled to antennas **417-1** and **417-2**. In the FIG. **4** example, the wireless network interface **416** is depicted as a 5G New Radio (NR)/WiFi 6 module, though this is not a requirement. The WOOBM device **410** further includes a System on Chip (SoC) **412** including a Central Processing Unit (CPU) core **415** and a gigabit ethernet controller **417** that is coupled to the ethernet port **418-1-1** via a physical layer circuit (PHY) **411**. The CPU core **415** has a Peripheral Component Interconnect Express (PCIe) (e.g., 1 lane) link to the gigabit ethernet controller **417**, as well as a PCIe (e.g., 1 lane) link to the wireless network interface **416**. Here, the ethernet port **418-1-1** is assumed to be PoE capable, such that once the WOOBM device **410** is connected to a PoE port of an IT asset (e.g., managed switch **206**, **306**, possibly via WOOBM adapters **219**, **319**), the PoE port may energize the WOOBM device **410** and cause it to act as a class 1 powered device. This may be implemented through a power distribution (PD) block **413**, which can provide suitable amounts of power to the SoC **412** and the wireless network interface **416**. Although 1G/full-duplex has become the management port standard, a PHY (e.g., **411**) may be used, for example, to adjust to sub-1 G speeds at the managed switch end or at the wireless network interface **416** end of the WOOBM device **410**.

(55) The gigabit ethernet controller **417** may be an integrated multiport device (e.g., a Media Access Control (MAC) switch) that has a first port (e.g., a managed interface port) that is internally managed by the SoC **412**. From the perspective of the CPU core **415**, the managed interface port may be a PCIe ethernet port that is used to configure the gigabit ethernet controller **417**. The gigabit ethernet controller **417** may have an SDK/kernel driver that may be used to transmit and receive packets (e.g., ethernet frames) over such an internal interface. The CPU core **415** has a PCIe (e.g., 1 lane) link to the gigabit ethernet controller **417**.

(56) The gigabit ethernet controller **417** in some embodiments may detect ethernet packets that are destined for an IT support system (e.g., a management switch or other device in the IT support system). Another port of the gigabit ethernet controller **417** is coupled to the ethernet port **418-1-1**. In some embodiments, the gigabit ethernet control **417** may be used to implement bump-in-the-wire management port throttling (e.g., in scenarios where a gigabit ethernet port of a switch does not support management port throttling to control the amount of ethernet traffic). The gigabit ethernet controller **417** may implement a rule to drop packets if the rate of untagged packets destined for the wireless network interface **416** exceeds some threshold (e.g., 1000 packets per second).

(57) As shown in FIG. **4**, the SoC **412** comprises a Universal Asynchronous Receiver/Transmitter (UART) port that is coupled to the console port **418-1-2**. In some embodiments, the UART port may establish primary console communication using a management switch or other device in an IT support system, a client device, or the WOOBM device **410** itself, as the console server to terminate the reverse telnet console session and reformat/package console traffic into ethernet frames, and to inject the reformatted console traffic into a PCIe port coupled to the wireless network interface **416**.

(58) An additional UART or USB port may be used to establish debugging of the OS or other software of the WOOBM device **410** itself (e.g., using flash drive/debug console **418-1-3**). In some embodiments, scripts that may execute on the WOOBM device **410** (e.g., CLI commands) can be downloaded and stored to the SoC **412** in addition to or in lieu of having access to an external server. The WOOBM device **410** may enable buffering to facilitate the capturing of console logs (e.g., in scenarios where connectivity has been temporarily lost, or where no reverse telnet session has been established yet since no users have connected to the console yet). The SoC **412** may

further implement a Trusted Platform Module (TPM) **419** connected to the CPU core **415** via a low pin count (LPC) bus. The SoC **412** may be further coupled to flash memory **412** and/or dynamic random-access memory (DRAM) **423**. In addition, the SoC **412** may implement Bluetooth or BLE radio functionality for establishing wireless network connections with, for example, client devices (e.g., client devices **202**, **302**). The Bluetooth or BLE radio functionality enables a user (e.g., of client devices **202**, **302**) to pair a smartphone application (e.g., WOOBM application **203**, **303**) using Bluetooth or BLE (or via a USB cable to flash drive/debug console **218-1-3**, **318-1-3** or a separate USB port not shown) to configure the WOOBM device **410** if desired (e.g., to set a static IP address, preferred 5G or other wireless network, download files, update the WOOBM device **410** boot image, debug the WOOBM device **410**, etc.).

(59) In some embodiments, the ethernet port **418-1-1** and gigabit ethernet controller **417** may operate at hardware speed or line rate with no significant delay (e.g., when performing multiplexing operations). In software implementations, various functions (e.g., encapsulation, decapsulation, mapping, etc.) may take advantage of one or more hardware accelerators (not shown) which may be implemented within the CPU core **415**.

(60) Data traffic through the WOOBM device **410** will now be described. In some embodiments, management port traffic ingresses on the ethernet port **418-1-1** (e.g., from a managed switch or other IT asset, not shown). In operation, the management port traffic is received by the gigabit ethernet controller **417**, and is then switched towards the CPU core **415** via the PCIe interface. The management port traffic may then be provided from the CPU core **415** to the wireless network interface **416** via another PCIe interface, possibly after various preprocessing by the CPU core **415**. The CPU core **415** also receives console port traffic that ingresses on the console port **418-1-2** (e.g., from a managed switch or other IT asset, not shown). Such console port traffic may then be provided from the CPU core **415** to the wireless network interface **416** via the PCIe interface, possible after various preprocessing by the CPU core **415**. The wireless network interface **416** provides such data traffic over one or more wireless networks using the antennas **417-1** and **417-2**.

(61) The antennas **417-1** and **417-2** may also receive data traffic that is destined for the WOOBM device **410** or an IT asset (e.g., a managed switch) that the WOOBM device **410** manages. Any received data traffic may be provided from the wireless network interface **416** to the CPU core **415**, and then may be switched to the ethernet port **418-1-1** and/or console port **418-1-2** as needed.

(62) Data traffic that is routed through the WOOBM device **410** may be switched via virtual local area networks (VLANs), which may be labeled as VLAN E (e.g., for management ethernet traffic to/from ethernet port **418-1-1**) and VLAN C (e.g., for console traffic to/from console port **418-1-2**). Incoming console traffic from the console port **418-1-2** may include console output data characters, which may be encapsulated by the CPU core **415** into VLAN-tagged packets or ethernet frames. It should be noted that such encapsulation or encoding may include any desired type or types of embedding, reformatting and packaging. The format of a serial output packet may comprise an ethernet header that (1) identifies the MAC address of a management switch or other device in an IT support system as the destination MAC address, (2) identifies the MAC address of the WOOBM device **410** as the source MAC address, and (3) tags the packet with a VLAN “C” tag to obtain a VLAN C packet. The VLAN C packets may be identified as console output and mapped to a reverse telnet console session (e.g., for access to the console of the IT asset or managed switch that the WOOBM device **410** is coupled to). Console and ethernet traffic (e.g., tagged with VLAN C and VLAN E, respectively), may be assigned different relative priority based on user preference.

(63) Console input traffic flow (e.g., received at the wireless network interface **416** from an IT support system or client device) may include VLAN C tagged console packets that include serial input data characters that result from a user establishing a reverse telnet session and inputting via keystrokes at a CLI. The reverse telnet session may convert such characters to ethernet frames and feed the frames over one or more wireless networks (e.g., as VLAN C tagged console packets having the MAC address of the WOOBM device **412** as the destination MAC address). The CPU

core **415** may decapsulate such ethernet frames to extract the characters and inject the decapsulated frames into the console port **418-1-2** via the UART port, such that the characters will eventually reach a console port of an IT asset (e.g., a managed switch) being managed by the WOOBM device **410**). It should be noted that the VLAN "E" and "C" tags may be hardcoded or assigned using various link discovery protocols.

(64) An initialization sequence for the WOOBM device **410** will now be described. Once the SoC **412** is energized and boots up, it may perform the initialization sequence which includes initializing UART, gigabit ethernet and wireless network interface (e.g., 5G/WiFi 6) drivers. The SoC **412** starts a DHCP client on the wireless network interface **416** (e.g., a wireless port thereof) and receives one or more IP addresses from a DHCP service (e.g., an Evolved Packet Core (EPC)) which are assigned to the WOOBM device **410** itself. The SoC **412** also starts a telnet or secure shell (SSH) service, which enables telnet or SSH connections to be established to an OS running on the WOOBM device **410** (e.g., to facilitate management and/or debugging operations, if desired). The SoC **412** may start a DHCP relay agent, so that the IP stack of the IT asset (e.g., the network switch) being managed can get one or more IP addresses assigned.

(65) For ethernet traffic handling, in some embodiments only packets with the IP address of the WOOBM device **410** set as the destination address are lifted by the WOOBM's IP stack—other packets are bridged between the wireless network interface **416** and the ethernet port **418-1-1** of the WOOBM device **410**. For console traffic handling, a reverse telnet session is started on the WOOBM device **410**'s IP address, with console traffic being bridged between the reverse telnet session and the UART port.

(66) In some cases, it may not be possible to provide or enable PoE for the management port of a WOOBM device. In these and other cases, an alternative design for the WOOBM device may be implemented which utilizes USB power where it is possible to do so. FIG. **5** shows a system **500** including a WOOBM device **510** with an ethernet port **518-1-1** and serial port **518-1-2** which are horizontally aligned, for use with a network switch **506** (also referred to as managed switch **506**) with a management ethernet port **518-2-1** (also referred to as management port **518-2-1**_ and console port **518-2-2** that are horizontally aligned. Here, an optional WOOBM connection port adapter **519** (also referred to as WOOBM adapter **519**) is shown, which also has horizontally aligned ethernet connectors **518-3-1** and serial connectors **518-3-2** for coupling the WOOBM device **510** to the managed switch **506** in a manner similar to that described above with respect to FIG. **2**. The WOOBM device **510** also includes a flash drive/debug console **518-1-3**, as well as antennas **517** for connecting with an IT support system **508** via RAN and NGCN **504** similar to the WOOBM device **210** of FIG. **2**. The WOOBM device **510** further includes a USB in port **518-1-4**, which is configured for coupling with a USB cable to a USB port **518-2-3** of the managed switch **506** for USB powering of the WOOBM device **510**. Here, a USB cable is used since an adapter-based solution may not be possible as the USB port **518-2-3** of the managed switch **506** is not collocated with the OOBM ports (e.g., management port **518-2-1** and console port **518-2-2**) due to the intervening stacking and system light-emitting diode (LED) indicators on the managed switch **506**. If the USB port **518-2-3** were collocated with the OOBM ports on the managed switch **506**, however, a WOOBM adapter **519** with integrated USB connectors may be used. An example of such an arrangement is shown in FIG. **6**.

(67) FIG. **6** shows a system **600** including a WOOBM device **610** with an ethernet port **618-1-1** and serial port **618-1-2** which are vertically aligned, for use with a network switch **606** (also referred to as managed switch **606**) with a management ethernet port **618-2-1** (also referred to as management port **618-2-1**) and console port **618-2-2** that are vertically aligned. Here, an optional WOOBM connection port adapter **619** (also referred to as WOOBM adapter **619**) is shown, which also has vertically aligned ethernet connectors **618-3-1** and serial connectors **618-3-2** for coupling the WOOBM device **610** to the managed switch **606** in a manner similar to that described above with respect to FIG. **3**. The WOOBM device **610** also includes a flash drive/debug console **618-1-3**, as

well as antennas **617** for connecting with an IT support system **608** via RAN and NGCN **604** similar to the WOOBM device **310** of FIG. **3**. The WOOBM device **610** further includes a USB in port **618-1-4**, which is configured for coupling via USB connectors **618-3-3** on the WOOBM adapter **619** to a USB port **618-2-3** of the managed switch **606** for USB powering of the WOOBM device **610**. Here, the adapter-based solution is possible as the USB port **618-2-3** of the managed switch **606** is collocated with the OOBM ports (e.g., management port **618-2-1** and console port **618-2-2**). The WOOBM device **610** also includes another optional USB in port **618-1-5** which may be used in a manner similar to that of the USB in port **518-1-4** for coupling with a USB cable.

(68) FIG. **7** shows a functional block diagram of a WOOBM device **710** (which may be the horizontally aligned WOOBM device **510** or the vertically aligned WOOBM device **610** that is USB powered). The WOOBM device **710** comprises an ethernet port **718-1-1** and console port **418-1-2**, both depicted as RJ45 ports, as well as USB ports **718-1-4** and **718-1-5** and a wireless network interface **716** coupled to antennas **717-1** and **717-2**. In the FIG. **7** example, the wireless network interface **716** is depicted as a 5G NR/WiFi 6 module, though this is not a requirement. The WOOBM device **710** further includes a SoC **712** including a CPU core **715** and a gigabit ethernet controller **717** that is coupled to the ethernet port **718-1-1** via PHY **711**. The CPU core **715** has a PCIe (e.g., 1 lane) link to the gigabit ethernet controller **717**, as well as a PCIe (e.g., 1 lane) link to the wireless network interface **716**.

(69) Here, the ethernet port **718-1-1** is assumed to not be PoE capable, such that the WOOBM device **710** is USB powered via connection at the USB port **718-1-4** (or possibly the USB port **718-1-5**) once the WOOBM device **710** is connected to a USB port of an IT asset (e.g., managed switch **506**, **606**). The USB power (which may be 5 watts (W) for USB type-A or 15 W for USB type-C), may energize the WOOBM device **710** and cause it to act as a class 1 powered device. This may be implemented through a 3-port USB switch **713-1**, USB (target) flash emulation **713-2** in the SoC **712**, and power block **713-3**.

(70) The gigabit ethernet controller **717** may be an integrated multiport device (e.g., a MAC switch) that has a first port (e.g., a managed interface port) that is internally managed by the SoC **712**. From the perspective of the CPU core **715**, the managed interface port may be a PCIe ethernet port that is used to configure the gigabit ethernet controller **717**. The gigabit ethernet controller **717** may have an SDK/kernel driver that may be used to transmit and receive packets (e.g., ethernet frames) over such an internal interface. The CPU core **715** has a PCIe (e.g., 1 lane) link to the gigabit ethernet controller **717**.

(71) The gigabit ethernet controller **717** in some embodiments may detect ethernet packets that are destined for an IT support system (e.g., a management switch or other device in the IT support system). Another port of the gigabit ethernet controller **717** is coupled to the ethernet port **718-1-1**. In some embodiments, the gigabit ethernet control **717** may be used to implement bump-in-the-wire management port throttling (e.g., in scenarios where a gigabit ethernet port of a switch does not support management port throttling to control the amount of ethernet traffic). The gigabit ethernet controller **717** may implement a rule to drop packets if the rate of untagged packets destined for the wireless network interface **716** exceeds some threshold (e.g., 1000 packets per second).

(72) As shown in FIG. **7**, the SoC **712** comprises a UART port that is coupled to the console port **718-1-2**. In some embodiments, the UART port may establish primary console communication using a management switch or other device in an IT support system, a client device, or the WOOBM device **710** itself, as the console server to terminate the reverse telnet console session and reformat/package console traffic into ethernet frames, and to inject the reformatted console traffic into a PCIe port coupled to the wireless network interface **716**.

(73) An additional UART or USB port may be used to establish debugging of the OS or other software of the WOOBM device **710** itself (e.g., using flash drive/debug console **718-1-3**). In some embodiments, scripts that may execute on the WOOBM device **710** (e.g., CLI commands) can be

downloaded and stored to the SoC **712** in addition to or in lieu of having access to an external server. The WOOBM device **710** may enable buffering to facilitate the capturing of console logs (e.g., in scenarios where connectivity has been temporarily lost, or where no reverse telnet session has been established yet since no users have connected to the console yet). The SoC **712** may further implement a TPM **719** connected to the CPU core **715** via an LPC bus. The SoC **712** may be further coupled to flash memory **721** and/or DRAM **723**. In addition, the SoC **712** may implement Bluetooth or BLE radio functionality for establishing wireless network connections with, for example, client devices. The Bluetooth or BLE radio functionality enables a user to pair a smartphone application using Bluetooth or BLE (or via a USB cable to flash drive/debug console **518-1-3**, **618-1-3**) to configure the WOOBM device **710** if desired (e.g., to set a static IP address, preferred 5G or other wireless network, download files, update the WOOBM device **710** boot image, debug the WOOBM device **710**, etc.).

(74) In some embodiments, the ethernet port **718-1-1** and gigabit ethernet controller **717** may operate at hardware speed or line rate with no significant delay (e.g., when performing multiplexing operations). In software implementations, various functions (e.g., encapsulation, decapsulation, mapping, etc.) may take advantage of one or more hardware accelerators (not shown) which may be implemented within the CPU core **715**.

(75) Data traffic through the WOOBM device **710** will now be described. In some embodiments, management port traffic ingresses on the ethernet port **718-1-1** (e.g., from a managed switch or other IT asset, not shown). In operation, the management port traffic is received by the gigabit ethernet controller **717**, and is then switched towards the CPU core **715** via the PCIe interface. The management port traffic may then be provided from the CPU core **715** to the wireless network interface **716** via another PCIe interface, possibly after various preprocessing by the CPU core **715**. The CPU core **715** also receives console port traffic that ingresses on the console port **718-1-2** (e.g., from a managed switch or other IT asset, not shown). Such console port traffic may then be provided from the CPU core **715** to the wireless network interface **716** via the PCIe interface, possible after various preprocessing by the CPU core. The wireless network interface **716** provides such data traffic over one or more wireless networks using the antennas **717-1** and **717-2**.

(76) The antennas **717-1** and **717-2** may also receive data traffic that is destined for the WOOBM device **710** or an IT asset (e.g., a managed switch) that the WOOBM device **710** manages. Any received data traffic may be provided from the wireless network interface **716** to the CPU core **715**, and then may be switched to the ethernet port **718-1-1** and/or console port **718-1-2** as needed.

(77) Data traffic that is routed through the WOOBM device **710** may be switched via VLANs, which may be labeled as VLAN E (e.g., for ethernet traffic to/from ethernet port **718-1-1**) and VLAN C (e.g., for console traffic to/from console port **718-1-2**). Incoming console traffic from the console port **718-1-2** may include console output data characters, which may be encapsulated by the CPU core **715** into VLAN-tagged packets or ethernet frames. It should be noted that such encapsulation or encoding may include any desired type or types of embedding, reformatting and packaging. The format of a serial output packet may comprise an ethernet header that (1) identifies the MAC address of a management switch or other device in an IT support system as the destination MAC address, (2) identifies the MAC address of the WOOBM device **710** as the source MAC address, and (3) tags the packet with a VLAN “C” tag to obtain a VLAN C packet. The VLAN C packets may be identified as console output and mapped to a reverse telnet console session (e.g., for access the console of the IT asset or managed switch that the WOOBM device **710** is coupled to). Console and ethernet traffic (e.g., tagged with VLAN C and VLAN E, respectively), may be assigned different relative priority based on user preference.

(78) Console input traffic flow (e.g., received at the wireless network interface **716** from an IT support system or client device) may include VLAN C tagged console packets that include serial input data characters that result from a user establishing a reverse telnet session and inputting via keystrokes at a CLI. The reverse telnet session may convert such characters to ethernet frames and

feed the frames over one or more wireless networks (e.g., as VLAN C tagged console packets having the MAC address of the WOOBM device **412** as the destination MAC address). The CPU core **715** may decapsulate such ethernet frames to extract the characters and inject the decapsulated frames into the console port **718-1-2** via the UART port, such that the characters will eventually reach a console port of an IT asset (e.g., a managed switch) being managed by the WOOBM device **710**). It should be noted that the VLAN "E" and "C" tags may be hardcoded or assigned using various link discovery protocols.

(79) An initialization sequence for the WOOBM device **710** will now be described. Once the SoC **712** is energized and boots up, it may perform the initialization sequence which includes initializing UART, gigabit ethernet and wireless network interface (e.g., 5G/WiFi 6) drivers. The SoC **712** starts a DHCP client on the wireless network interface **716** (e.g., a wireless port thereof) and receives one or more IP addresses from a DHCP service (e.g., an EPC) which are assigned to the WOOBM device **710** itself. The SoC **712** also starts a telnet or SSH service, which enables telnet or SSH connections to be established to an OS running on the WOOBM device **710** (e.g., to facilitate management and/or debugging operations, if desired). The SoC **712** may start a DHCP relay agent, so that the IP stack of the IT asset (e.g., the network switch) being managed can get one or more IP addresses assigned.

(80) For ethernet traffic handling, only packets with the IP address of the WOOBM device **710** set as the destination address are lifted by the WOOBM's IP stack—other packets are bridged between the wireless network interface **716** and the ethernet port **718-1-1** of the WOOBM device **710**. For console traffic handling, a reverse telnet session is started on the WOOBM device **710**'s IP address, with console traffic being bridged between the reverse telnet session and the UART port.

(81) An exemplary process for WOOBM of IT assets will now be described in more detail with reference to the flow diagram of FIG. **8**. It is to be understood that this particular process is only an example, and that additional or alternative processes for WOOBM of IT assets may be used in other embodiments.

(82) In this embodiment, the process includes steps **800** through **804**. These steps are assumed to be performed by the WOOBM device **110** utilizing the OOBM traffic management logic **120**. The process begins with step **800**, receiving, utilizing a first one of the connection ports **118-1**, a first type of OOBM traffic from managed IT asset **106**. In step **802**, a second type of OOBM traffic from the managed IT asset **106** is received utilizing a second one of the connection ports **118-1**. The first connection port may comprise a serial port, and the first type of OOBM traffic may comprise console traffic from the managed IT asset **106**. The second connection port may comprise an ethernet port, and the second type of OOBM traffic may comprise ethernet traffic from the managed IT asset **106**. The managed IT asset **106** may comprise a network switch. In some embodiments, the WOOBM device **110** receives power at the ethernet port from a PoE port of the managed IT asset **106**. In other embodiments, the WOOBM device **110** receives power at a third one of the connection ports **118-1** from the managed IT asset **106**. The third connection port may comprise a USB port. In some embodiments, a connection port adapter (e.g., WOOBM connection port adapter **119**) may be used to couple the first and second connection ports to corresponding connection ports **118-2** of the managed IT asset **106**.

(83) The WOOBM device **110** in step **804** communicates, via the wireless network interfaces **116**, the first and second types of OOBM traffic to one or more additional processing devices (e.g., client device **102**, IT support system **108**). In some embodiments, step **804** may comprise multiplexing the first and second types of OOBM traffic into one or more ethernet packets, and communicating the one or more ethernet packets comprising the multiplexed OOBM traffic. The at least one wireless network interface may utilize a cellular wireless network for communicating the first and second types of OOBM traffic to at least one of the one or more additional processing devices. The cellular wireless network may comprise a private 5G network of an enterprise providing support services for the managed IT asset **106** utilizing said at least one of the one or

more additional processing devices. The at least one wireless network interface may also or alternatively utilize a WiFi wireless network for communicating the first and second types of OOBM traffic to at least one of the one or more additional processing devices. The at least one wireless network interface may also or alternatively utilize at least one of a Bluetooth and a BLE network for communicating the first and second types of OOBM traffic to at least one of the one or more additional processing devices. At least one of the one or more additional processing devices may be part of a cloud-based IT support system (e.g., IT support system **108**) responsible for providing support services for the managed IT asset **106**.

(84) In some embodiments, the FIG. **8** process may further include receiving, via the wireless network interfaces **116**, additional OOBM traffic from at least one of the one or more additional processing devices and separating the additional OOBM traffic into the first type of OOBM traffic and the second type of OOBM traffic. The first type of OOBM traffic in the additional OOBM traffic is communicated utilizing the first connection port to a first one of the OOBM connection ports **118-2** of the managed IT asset **106**, and the second type of OOBM traffic in the additional OOBM traffic is communicated utilizing the second connection port to a second one of the OOBM connection ports **118-2** of the managed IT asset **106**.

(85) The technical solutions described herein provide WOOBM devices or dongles (e.g., vertically or horizontally aligned) that are configured for connection to management and console ports of an IT asset being managed (e.g., a managed network switch), possibly with the use of WOOBM adapters, to multiplex management and console traffic from the IT asset being managed over one or more wireless networks (e.g., cellular networks such as LTE or 5G networks, WiFi networks, etc.). The technical solutions described herein thus advantageously enable ease of configuration and control of the IT asset being managed, reduced cabling, and WOOBM-based value added services (e.g., where a network operator or other user with a smartphone or other type of client device may interact with the WOOBM devices using Bluetooth or another suitable wireless communication protocol). Conventional WiFi or LTE-based USB adapters do not address the use case for networking switches and other types of IT assets where console and management (e.g., ethernet) traffic is multiplexed over a wireless interface. Further, conventional approaches do not enable access to multiple OOBM ports (e.g., management/ethernet and console ports) of an IT asset with a single wireless interface.

(86) The WOOBM devices described herein may be used for various telecommunications, multi-cloud, enterprise and edge solutions where IT assets (e.g., network switches) are deployed at 5G cell sites, as well as enterprises that are deploying private 5G. The WOOBM devices described herein may be built, tested and eventually bundled with the IT assets they are designed to manage (e.g., for customers or other end-users who are planning to migrate management networks to LTE/5G or WiFi 6). Wireless wide area network (WWAN) vendors may also use WOOBM devices described herein. As described above, the WOOBM devices may be configured for powering via PoE (e.g., for IT assets such as network switches whose management ports are PoE-capable) or via USB (e.g., for IT assets such as network switches whose management ports are not PoE-capable).

(87) It is to be appreciated that the particular advantages described above and elsewhere herein are associated with particular illustrative embodiments and need not be present in other embodiments. Also, the particular types of information processing system features and functionality as illustrated in the drawings and described above are exemplary only, and numerous other arrangements may be used in other embodiments.

(88) Illustrative embodiments of processing platforms utilized to implement functionality for WOOBM of IT assets will now be described in greater detail with reference to FIGS. **9** and **10**. Although described in the context of system **100**, these platforms may also be used to implement at least portions of other information processing systems in other embodiments.

(89) FIG. **9** shows an example processing platform comprising cloud infrastructure **900**. The cloud infrastructure **900** comprises a combination of physical and virtual processing resources that may

be utilized to implement at least a portion of the information processing system **100** in FIG. **1**. The cloud infrastructure **900** comprises multiple virtual machines (VMs) and/or container sets **902-1**, **902-2**, . . . **902-**L implemented using virtualization infrastructure **904**. The virtualization infrastructure **904** runs on physical infrastructure **905**, and illustratively comprises one or more hypervisors and/or operating system level virtualization infrastructure. The operating system level virtualization infrastructure illustratively comprises kernel control groups of a Linux operating system or other type of operating system.

(90) The cloud infrastructure **900** further comprises sets of applications **910-1**, **910-2**, . . . **910-**L running on respective ones of the VMs/container sets **902-1**, **902-2**, . . . **902-**L under the control of the virtualization infrastructure **904**. The VMs/container sets **902** may comprise respective VMs, respective sets of one or more containers, or respective sets of one or more containers running in VMs.

(91) In some implementations of the FIG. **9** embodiment, the VMs/container sets **902** comprise respective VMs implemented using virtualization infrastructure **904** that comprises at least one hypervisor. A hypervisor platform may be used to implement a hypervisor within the virtualization infrastructure **904**, where the hypervisor platform has an associated virtual infrastructure management system. The underlying physical machines may comprise one or more distributed processing platforms that include one or more storage systems.

(92) In other implementations of the FIG. **9** embodiment, the VMs/container sets **902** comprise respective containers implemented using virtualization infrastructure **904** that provides operating system level virtualization functionality, such as support for Docker containers running on bare metal hosts, or Docker containers running on VMs. The containers are illustratively implemented using respective kernel control groups of the operating system.

(93) As is apparent from the above, one or more of the processing modules or other components of system **100** may each run on a computer, server, storage device or other processing platform element. A given such element may be viewed as an example of what is more generally referred to herein as a “processing device.” The cloud infrastructure **900** shown in FIG. **9** may represent at least a portion of one processing platform. Another example of such a processing platform is processing platform **1000** shown in FIG. **10**.

(94) The processing platform **1000** in this embodiment comprises a portion of system **100** and includes a plurality of processing devices, denoted **1002-1**, **1002-2**, **1002-3**, . . . **1002-**K, which communicate with one another over a network **1004**.

(95) The network **1004** may comprise any type of network, including by way of example a global computer network such as the Internet, a WAN, a LAN, a satellite network, a telephone or cable network, a cellular network, a wireless network such as a WiFi or WiMAX network, or various portions or combinations of these and other types of networks.

(96) The processing device **1002-1** in the processing platform **1000** comprises a processor **1010** coupled to a memory **1012**.

(97) The processor **1010** may comprise a microprocessor, a microcontroller, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a central processing unit (CPU), a graphical processing unit (GPU), a tensor processing unit (TPU), a video processing unit (VPU) or other type of processing circuitry, as well as portions or combinations of such circuitry elements.

(98) The memory **1012** may comprise random access memory (RAM), read-only memory (ROM), flash memory or other types of memory, in any combination. The memory **1012** and other memories disclosed herein should be viewed as illustrative examples of what are more generally referred to as “processor-readable storage media” storing executable program code of one or more software programs.

(99) Articles of manufacture comprising such processor-readable storage media are considered illustrative embodiments. A given such article of manufacture may comprise, for example, a storage

array, a storage disk or an integrated circuit containing RAM, ROM, flash memory or other electronic memory, or any of a wide variety of other types of computer program products. The term “article of manufacture” as used herein should be understood to exclude transitory, propagating signals. Numerous other types of computer program products comprising processor-readable storage media can be used.

(100) Also included in the processing device **1002-1** is network interface circuitry **1014**, which is used to interface the processing device with the network **1004** and other system components, and may comprise conventional transceivers.

(101) The other processing devices **1002** of the processing platform **1000** are assumed to be configured in a manner similar to that shown for processing device **1002-1** in the figure.

(102) Again, the particular processing platform **1000** shown in the figure is presented by way of example only, and system **100** may include additional or alternative processing platforms, as well as numerous distinct processing platforms in any combination, with each such platform comprising one or more computers, servers, storage devices or other processing devices.

(103) For example, other processing platforms used to implement illustrative embodiments can comprise converged infrastructure.

(104) It should therefore be understood that in other embodiments different arrangements of additional or alternative elements may be used. At least a subset of these elements may be collectively implemented on a common processing platform, or each such element may be implemented on a separate processing platform.

(105) As indicated previously, components of an information processing system as disclosed herein can be implemented at least in part in the form of one or more software programs stored in memory and executed by a processor of a processing device. For example, at least portions of the functionality for WOOBM of IT assets as disclosed herein are illustratively implemented in the form of software running on one or more processing devices.

(106) It should again be emphasized that the above-described embodiments are presented for purposes of illustration only. Many variations and other alternative embodiments may be used. For example, the disclosed techniques are applicable to a wide variety of other types of information processing systems, IT assets, etc. Also, the particular configurations of system and device elements and associated processing operations illustratively shown in the drawings can be varied in other embodiments. Moreover, the various assumptions made above in the course of describing the illustrative embodiments should also be viewed as exemplary rather than as requirements or limitations of the disclosure. Numerous other alternative embodiments within the scope of the appended claims will be readily apparent to those skilled in the art.

Claims

1. An apparatus comprising: two or more connection ports; at least one wireless network interface; and at least one processing device comprising a processor coupled to a memory; the at least one processing device being configured: to receive, utilizing a first one of the two or more connection ports, a first type of out-of-band management traffic from a managed information technology asset; to receive, utilizing a second one of the two or more connection ports, a second type of out-of-band management traffic from the managed information technology asset; to communicate, via the at least one wireless network interface, the first and second types of out-of-band management traffic to one or more additional processing devices; to receive, via the at least one wireless network interface, additional out-of-band management traffic from at least one of the one or more additional processing devices; to determine whether the additional out-of-band management traffic includes any out-of-band management traffic of the first and second types of out-of-band management traffic; responsive to determining that at least a first portion of the additional out-of-band management traffic includes the first type of out-of-band management traffic, to communicate the

first portion of the additional out-of-band management traffic to the managed information technology asset utilizing the first connection port; and responsive to determining that at least a second portion of the additional out-of-band management traffic includes the second type of out-of-band management traffic, to communicate the second portion of the additional out-of-band management traffic to the managed information technology asset utilizing the second connection port.

2. The apparatus of claim 1 wherein the first connection port comprises a serial port and the first type of out-of-band management traffic comprises console traffic from the managed information technology asset.

3. The apparatus of claim 2 wherein the second connection port comprises an ethernet port and the second type of out-of-band management traffic comprises ethernet traffic from the managed information technology asset.

4. The apparatus of claim 1 wherein the managed information technology asset comprises a network switch.

5. The apparatus of claim 1 wherein the at least one processing device is further configured to receive power via at least one of the first connection port and the second connection port, said at least one of the first connection port and the second connection port comprising a power over ethernet port of the managed information technology asset.

6. The apparatus of claim 1 wherein the at least one processing device is further configured to receive power at a third one of the two or more connection ports from the managed information technology asset.

7. The apparatus of claim 6 wherein the third connection port comprises a universal serial bus port.

8. The apparatus of claim 1 wherein the at least one processing device is further configured to multiplex the first and second types of out-of-band management traffic into one or more ethernet packets, and wherein communicating the first and second types of out-of-band management traffic to the one or more additional processing devices comprises communicating the one or more ethernet packets comprising the multiplexed out-of-band management traffic.

9. The apparatus of claim 1 wherein the at least one wireless network interface utilizes a cellular wireless network for communicating the first and second types of out-of-band management traffic to at least one of the one or more additional processing devices.

10. The apparatus of claim 9 wherein the cellular wireless network comprises a private 5G network of an enterprise providing support services for the managed information technology asset utilizing said at least one of the one or more additional processing devices.

11. The apparatus of claim 1 wherein the at least one wireless network interface utilizes a WiFi wireless network for communicating the first and second types of out-of-band management traffic to at least one of the one or more additional processing devices.

12. The apparatus of claim 1 wherein the at least one wireless network interface utilizes at least one of a Bluetooth and a Bluetooth low energy network for communicating the first and second types of out-of-band management traffic to at least one of the one or more additional processing devices.

13. The apparatus of claim 1 wherein at least one of the one or more additional processing devices is part of a cloud-based information technology support system responsible for providing support services for the managed information technology asset.

14. The apparatus of claim 1 wherein the at least one processing device is further configured: to separate the additional out-of-band management traffic into the first type of out-of-band management traffic and the second type of out-of-band management traffic; to communicate, utilizing the first connection port, the first type of out-of-band management traffic in the additional out-of-band management traffic to a first out-of-band management port of the managed information technology asset; and to communicate, utilizing the second connection port, the second type of out-of-band management traffic in the additional out-of-band management traffic to a second out-of-band management port of the managed information technology asset.

15. A computer program product comprising a non-transitory processor-readable storage medium having stored therein program code of one or more software programs, wherein the program code when executed by at least one processing device causes the at least one processing device: to receive, utilizing a first one of two or more connection ports, a first type of out-of-band management traffic from a managed information technology asset; to receive, utilizing a second one of the two or more connection ports, a second type of out-of-band management traffic from the managed information technology asset; to communicate, via at least one wireless network interface, the first and second types of out-of-band management traffic to one or more additional processing devices; to receive, via the at least one wireless network interface, additional out-of-band management traffic from at least one of the one or more additional processing devices; to determine whether the additional out-of-band management traffic includes any out-of-band management traffic of the first and second types of out-of-band management traffic; responsive to determining that at least a first portion of the additional out-of-band management traffic includes the first type of out-of-band management traffic, to communicate the first portion of the additional out-of-band management traffic to the managed information technology asset utilizing the first connection port; and responsive to determining that at least a second portion of the additional out-of-band management traffic includes the second type of out-of-band management traffic, to communicate the second portion of the additional out-of-band management traffic to the managed information technology asset utilizing the second connection port.
16. The computer program product of claim 15 wherein the first connection port comprises a serial port, the first type of out-of-band management traffic comprises console traffic from the managed information technology asset, the second connection port comprises an ethernet port, and the second type of out-of-band management traffic comprises ethernet traffic from the managed information technology asset.
17. The computer program product of claim 15 wherein the managed information technology asset comprises a network switch.
18. A method performed by at least one processing device comprising a process coupled to a memory, the method comprising: receiving, utilizing a first one of two or more connection ports, a first type of out-of-band management traffic from a managed information technology asset; receiving, utilizing a second one of the two or more connection ports, a second type of out-of-band management traffic from the managed information technology asset; communicating, via at least one wireless network interface, the first and second types of out-of-band management traffic to one or more additional processing devices; receiving, via the at least one wireless network interface, additional out-of-band management traffic from at least one of the one or more additional processing devices; determining whether the additional out-of-band management traffic includes any out-of-band management traffic of the first and second types of out-of-band management traffic; responsive to determining that at least a first portion of the additional out-of-band management traffic includes the first type of out-of-band management traffic, communicating the first portion of the additional out-of-band management traffic to the managed information technology asset utilizing the first connection port; and responsive to determining that at least a second portion of the additional out-of-band management traffic includes the second type of out-of-band management traffic, communicating the second portion of the additional out-of-band management traffic to the managed information technology asset utilizing the second connection port.
19. The method of claim 18 wherein the first connection port comprises a serial port, the first type of out-of-band management traffic comprises console traffic from the managed information technology asset, the second connection port comprises an ethernet port, and the second type of out-of-band management traffic comprises ethernet traffic from the managed information technology asset.

20. The method of claim 18 wherein the managed information technology asset comprises a network switch.